Neural Nexus 2.0

Team Name : SYNTAX ERROR

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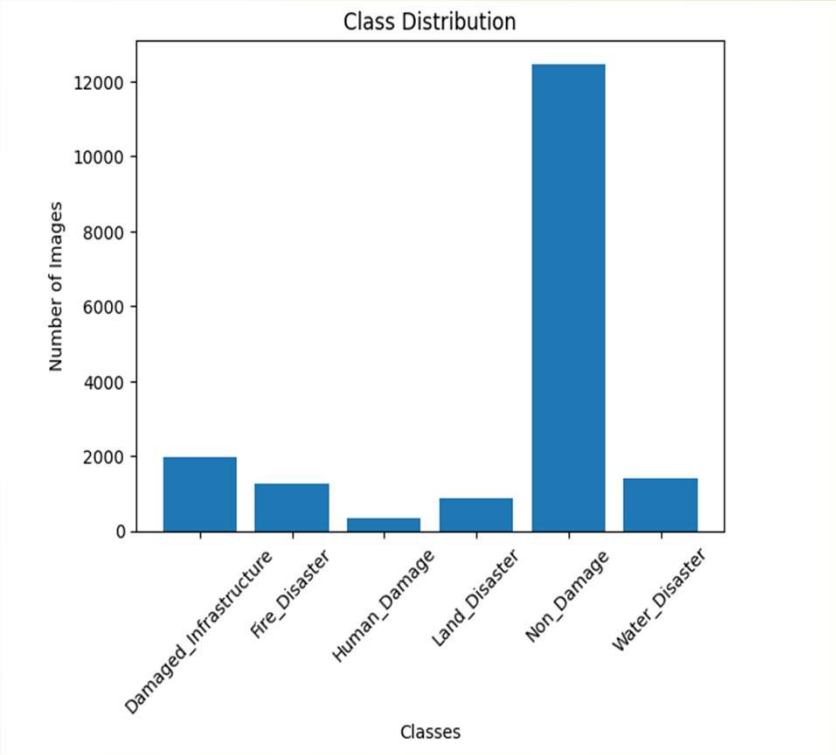
Problem Statement

Using the Disaster Images Dataset, design and develop an advanced *AI/ML-driven system* for fine-grained disaster scene understanding and impact assessment from real-world imagery. The objective is to build a robust and generalizable model capable of accurately classifying disaster types and extracting meaningful visual patterns that reflect severity and contextual damage, while addressing challenges such as high intra-class variability, inter-class similarity, noisy or ambiguous labels, class imbalance, and diverse environmental conditions (lighting, occlusions, perspectives). Participants are expected to implement effective preprocessing strategies, and explore advanced techniques such as *attention mechanisms, transfer learning,* and representation learning to capture both global scene context and localized damage cues. The solution should further emphasize robust generalization, interpretability, and scalability, enabling reliable performance on unseen data and supporting real-world applications in disaster monitoring, emergency response, and situational awareness systems.

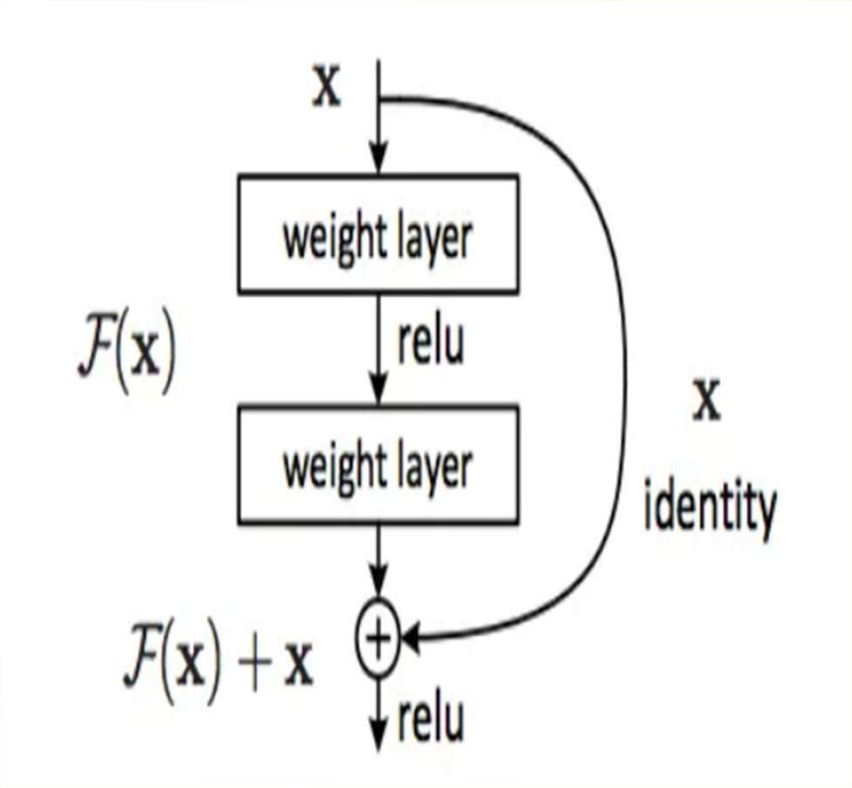
Data Preprocessing

1.Random resized crop (224x224)
2.Horizontal and vertical flips
3.Normalization (ImageNet mean/std)
4.Resize to 224x224
5.Normalization
6. Class Imbalance Handling:
Computed class distribution using label counts
Applied inverse frequency weighting
Used WeightedRandomSampler to balance training batches



Why did we use ResNet50 ?

•**Solving the Vanishing Gradient Problem:** As networks go deeper, gradients can disappear, making it difficult for earlier layers to learn. ResNet50 solves this by using "skip connections" (or "shortcuts") that allow gradients to flow directly through the network, easing optimization.
•**Optimal Depth for Performance:** With 50 layers, it is deep enough to learn complex features—such as intricate textures or shapes in medical imaging—but not so deep that it becomes impossible to train or prohibitively slow, unlike 101 or 152-layer variants.
•**Excellent for Transfer Learning:** ResNet50 is widely pre-trained on the ImageNet dataset. Using these pre-trained weights allows for fine-tuning on smaller or specialized datasets (such as medical MRIs or industrial defect detection) with limited data, leading to higher accuracy and faster convergence.
•**Bottleneck Block Structure:** ResNet50 uses a "bottleneck" block design, which reduces computational complexity by employing 1x1 convolutions to reduce the number of channels before processing and 1x1 convolutions to increase them back, making it more efficient than older networks like VGG.



• Why didn't we use other models ?

•**ResNet50 vs. VGG:** ResNet50 is deeper and faster to train.
•**ResNet50 vs. MobileNet:** MobileNet is better for mobile devices, but ResNet50 provides higher accuracy for standard computers.
•**ResNet50 vs. InceptionV3:** InceptionV3 focuses on width (using parallel filters of different sizes) while ResNet50 focuses on depth; Inception can achieve slightly higher accuracy on ImageNet but has a more complex architecture that is harder to modify.

Transfer Learning:

•**Load Pre-trained Weights:** Import a ResNet model initialized with ImageNet weights to act as a feature extractor. These early layers already recognize general patterns like edges, textures, and shapes.

Pre-trained Model: Start with a model already trained on a large dataset for a specific task. This pre-trained model has learned general features and patterns that are relevant across related tasks.

Base Model: This pre-trained model, known as the base model, includes layers that have processed data to learn hierarchical representations, capturing low-level to complex features.

Transfer Layers: Identify layers within the base model that hold generic information applicable to both the original and new tasks. Lower layers capture general features such as edges and textures, while higher layers capture task-specific complex patterns.

Fine-tuning: Fine-tune these selected layers with data from the new task. This process helps retain the pre-trained knowledge while adjusting parameters to meet the specific requirements of the new task, improving accuracy and adaptability.

Classification Metrics

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==================================================
  Overall Accuracy    : 0.9735
  Weighted Precision  : 0.9738
  Weighted Recall     : 0.9735
  Macro F1 Score      : 0.9335
  Weighted F1 Score   : 0.9736
==================================================

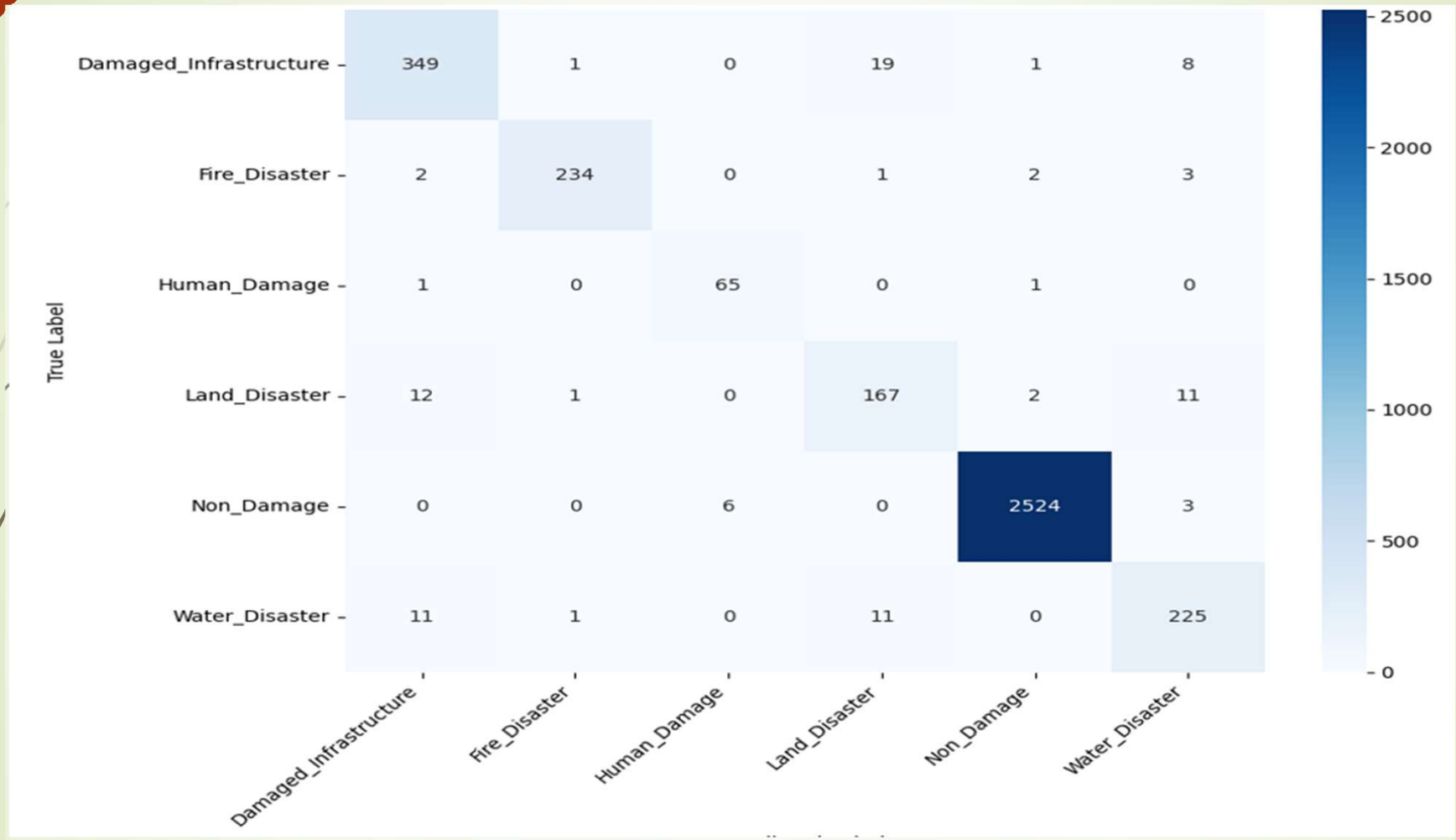
Classification Report:

                         precision    recall  f1-score   support

Damaged_Infrastructure        0.93      0.92      0.93       378
          Fire_Disaster       0.99      0.97      0.98       242
           Human_Damage       0.92      0.97      0.94        67
          Land_Disaster       0.84      0.87      0.85       193
             Non_Damage       1.00      1.00      1.00      2533
         Water_Disaster       0.90      0.91      0.90       248

               accuracy                           0.97      3661
              macro avg       0.93      0.94      0.93      3661
           weighted avg       0.97      0.97      0.97      3661
```

CONFUSION MATRIX



Innovation

Severity Classification: AI models can categorize imagery into levels such as "severe" (collapsed structures), "mild" (partial damage), and "little/none," often with high accuracy.

Rapid Damage Assessment: Automated systems that we created will process thousands of images much faster than manual inspection, which is critical in the first hours of a disaster.

Multi-Modal Analysis: Advanced systems combine visual data with textual information (e.g., social media posts saying "we are trapped") to increase the accuracy of determining the need for help.

Resource Allocation: By mapping high-severity areas, rescue teams can prioritize life-saving efforts and deploy resources more efficiently.